



AI-Based Multi-Task Image Enhancement Using Diffusion Models

Course: MS in Applied Artificial Intelligence AAI 521

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TECHNICAL REPORT

1. Introduction

This technical report presents the design, implementation, and evaluation of a multi-task image enhancement system built using diffusion models utilizing two approaches:

- Low-Rank Adaptation (LoRA) fine-tuning.
- Full Unet training

The project integrates four distinct image-processing tasks Colorization, Inpainting, Denoising, and Super Resolution into a unified web-based platform deployed on HuggingFace Spaces.

The motivation behind this system stems from real-world needs in photography restoration, digital media enhancement, medical imaging preprocessing, and archival reconstruction.

Traditional techniques for such tasks either lack realism or fail to generalize across diverse images. Diffusion models have emerged as state-of-the-art due to their iterative noise reconstruction and ability to generate highly realistic images.

The primary objectives of this project were:

- To retrain the Unet networks on specific tasks
 - To fine-tune diffusion models using LoRA to efficiently support multiple enhancement tasks
 - To create an accessible web interface for non-technical users
 - To build modular backend architecture that enables scalable deployment
 - To evaluate enhanced models against base diffusion pipelines qualitatively
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2. Background and Literature Review

2.1 Diffusion Models

Diffusion models generate data by denoising random noise through multiple iterative steps.

Their strengths include:

- High fidelity
- Stability compared to GANs
- Strong semantic reasoning

Stable Diffusion, a latent open source diffusion model, enables efficient training and inference by operating in a compressed latent space.

2.2 Image Colorization

Colorization historically relied on optimization or patch-based blending. Deep learning introduced CNN semantic colorization. Diffusion-based colorization yields more coherent color tones, especially for regions requiring semantic understanding (sky, foliage, skin tones).

2.3 Image Inpainting

Classical methods (Telea, Criminisi) fill missing regions using local information but fail for complex textures.

GAN-based approaches improve realism but suffer from mode collapse and structural inconsistency.

Diffusion inpainting pipelines enable:

- Mask-based conditioning
- Gradual restoration
- Texture-coherent filling

StableDiffusionInpaintPipeline is currently among the strongest general-purpose inpainting models.

2.4 Denoising

Image denoising seeks to recover clean images from noise-corrupted observations. Early approaches used Gaussian filters, which smoothed images indiscriminately, causing loss of edges and details (Tomasi & Manduchi, 1998). Edge-preserving methods like bilateral filtering addressed this by weighting pixels based on both spatial proximity and intensity similarity, though they could introduce artifacts such as staircase effects (Paris et al., 2009).

Deep learning transformed denoising performance. Zhang et al. (2017) introduced DnCNN, which used residual learning to predict noise directly rather than clean images, consistently outperforming classical methods like BM3D across noise levels. UNet architectures with skip connections further improved multi-scale feature integration (Ronneberger et al., 2015).

See **Appendix B** For further background information

2.6 Low-Rank Adaptation (LoRA)

LoRA reduces fine-tuning overhead by training a small set of low-rank adapters inserted into attention layers.

Advantages:

- As low as 99% fewer trainable parameters
- Faster training
- Works well for diffusion models
- Easy merging/unmerging for deployment

We used LoRA for both Colorization and Inpainting models.

2.7 Full UNet Training as an Alternative to

LoRA

While Low-Rank Adaptation (LoRA) provides an efficient mechanism for adapting diffusion models by updating only small low rank adapter matrices inserted into attention layers (Hu et al., 2022), an alternative approach is to fine-tune the **entire UNet backbone** directly. Full model training has significantly higher computational and memory costs, but can offer benefits in scenarios where LoRA's representational constraints limit its performance.

In diffusion models, the **UNet** serves as the denoising function that predicts ϵ (noise) at each timestep (Ho et al., 2020). It contains convolutional blocks, cross-attention modules, residual connections, and up/downsampling paths that collectively learn multi-scale structure.

Fine-tuning the entire UNet updates *all* of these components including convolutional filters, attention projections, and timestep embeddings, allowing the model to fully adapt to the statistics of the target domain.

Use Cases in This Project

In our work, LoRA was used for colorization and inpainting due to efficiency and ease of deployment. However, we explored **full UNet training** for the denoiser and super-resolution tasks when dealing with more challenging forms of image degradation, where the model required deeper architectural adaptation.

3. Dataset

We use the **COCO (Common Objects in Context)** dataset, a large-scale benchmark with diverse scenes.

Subset Used

- 3,000 images
- Variety of objects, lighting conditions, semantic diversity
- High-quality segmentation masks used for inpainting tasks

Preprocessing

- Resize to 512×512
- Convert to grayscale for colorization images
- Generate random rectangular masked regions for inpainting images
- Create binary mask for inpainting pipeline inference
- Convert mask to white color for inpainting images
- Add noise for denoising

- Random crops/augmentations for robustness
-

4. System Architecture

4.2 Components

A. Frontend: HuggingFace Gradio UI

The interface written in Gradio and includes:

- Image upload panel
- Operation selector dropdown
- Output display panel
- “Run” button that triggers inference

This simple layout enables quick experimentation.

C. Models

- **StableDiffusionPipeline** for colorization
- **StableDiffusionInpaintPipeline** for inpainting
- **StableDiffusionImg2ImgPipeline** for noise removal
- **StableDiffusionUpscalePipeline** for super scaling/resolution

5. Implementation Details

5.1 Training Setup

- LoRA rank: 8
- Learning rate: $1e-4$
- Batch size: 4
- Optimizer: AdamW
- Epochs: 5–9 (limited due to runtime interruptions)
- Hardware: Google Colab GPU + offline compute

5.2 Inference Pipeline

Input → Preprocessing → Pipeline Execution → Post-Processing → Output

- Preprocessing: Convert to tensor, grayscale/mask generation
- Inference: 20–30 diffusion steps
- Post-processing: color correction, merging masked regions

5.3 Training Denoiser Implementation details

5.3.1 Fine-Tuning the U-Net for Diffusion-Based Image Denoising

To develop a high-performance denoising model, we fine-tuned the **U-Net denoiser component** within the *Stable Diffusion Img2Img* pipeline. Stable Diffusion is a latent diffusion model (LDM) that performs image generation and transformation in a compressed latent space using a combination of a **Variational Autoencoder (VAE)**, a **U-Net noise predictor**, and a **text encoder** for conditioning (Rombach et al., 2022). Since denoising quality is governed primarily by the U-Net’s ability to estimate and remove noise, it serves as the core trainable component in our system. See **appendix D** for further implementation details.

See Appendix G for Metrics Details

5.4 Fine-Tuning the U-Net for Diffusion-Based Super-Resolution

To train a high quality super-resolution (SR) model, we fine-tuned the **U-Net of the Stable Diffusion Upscaler pipeline**, a latent diffusion model designed for image enhancement tasks. Stable Diffusion Upscaler operates by conditioning the U-Net on both a low-resolution input image and high-level text embeddings, allowing it to reconstruct high-frequency details consistent with the scene content (Rombach et al., 2022). Since the U-Net is responsible for predicting the noise removed during reverse diffusion, it serves as the primary component determining SR quality. See **appendix D** for further implementation details.

6. Results

6.1 Colorizer and Inpainting Qualitative Results

Colorization

- Base model and pipeline produced similar results to the fine tuned model pipeline
- Both models struggle slightly with maintaining original image structure while adding colors
- Colorization produced colorized results consistently
- Adjustments to prompt, strength and guidance_scale parameters allowed for improved control of maintaining original image structure while producing consistent color images (Lin et al., 2023)

Inpainting

- LoRA model consistently infilled missing regions for the masked images
- Both base and fine tuned model pipelines had similar quality image output
- Artifacts were present in darker image infill tasks, but images with better lighting seemed to produce less artifacts and maintain original image structure

Denoising Results

For the denoising results we captured the PSNR, SSIM and LPIPS for the standard stable diffusion model and the fine tuned diffusion model via the approximately 860 million trainable parameters or U-Net weights.

Table 1 shows that there metrics seem to be improved over the base model however visually inspecting the results shows that there is a lot of room for improvement. Furthermore there may be an underlying issue or bug in how the pipeline is fine tuning the standard model even though the metrics are improving.

Table 1: Denoising performance

Performance Metric	Test Set SD v1.5	Test Set UNet trained (ours)
PSNR	12.7420	15.7459
SSIM	0.1270	0.1540
LPIPS	0.7009	0.6604

See **Appendix F** Figure 3 for Results Images.

It seems that there was minimal improvement over the 3000 steps it was run for. Future work here would be to incorporate the training loss approach taken in the inpainting training loop which shows clear elbow points as training progresses versus the noisy evaluation at the end of each step.

6.2 Super-Resolution Results

For super resolution it was not clear on how to apply the metrics since the image was upscaled from the original size. One idea was to downsample the upscaled result back to the original size and then compare that pixel to pixel with the original image. This however shows a large difference even if the new image was far superior to the original due to improved details and edges. Human review was deemed the best way to evaluate the results from super scaling with the downside that we could not see improving metrics as training progressed.

The super-resolution experiments compared two models: the low-resolution input, the pretrained Stable Diffusion v1.5 upscaler, and retrained super-resolution (SR) UNet model which comprised approx 860 million trainable parameters. The qualitative results, shown in **Figure 5**, reveal that while the pretrained SD v1.5 upscaler offers modest improvements over the blurry inputs, the trained/fine-tuned SR model did not achieve the expected level of detail restoration due to limited training epochs. See **Appendix F** for further result analysis.

6.2 Colorizer Quantitative Results

In addition to MSE Loss, PSNR, SSI, Frechet Inception Distance (FID) was also measured for the Colorizer performance to measure feature distributions across images compared to true image features. A LoRA rank of 8 was used for training a UNet model for the

StableDiffusionImg2ImgPipeline, resulting in 1,594,368 of 861,115,332 (0.18%) trainable parameters. This appeared to be far too low as performance metrics appeared to stabilize after 2 epochs of training as shown in Figure 6. This implies that training is not very effective for the small amount of weights being adjusted and may also explain the reason for the similar base model and fine-tuned (FT) model performance metrics similarity on the test set shown in Table 2.

Figure 6: Colorizer performance metrics during training

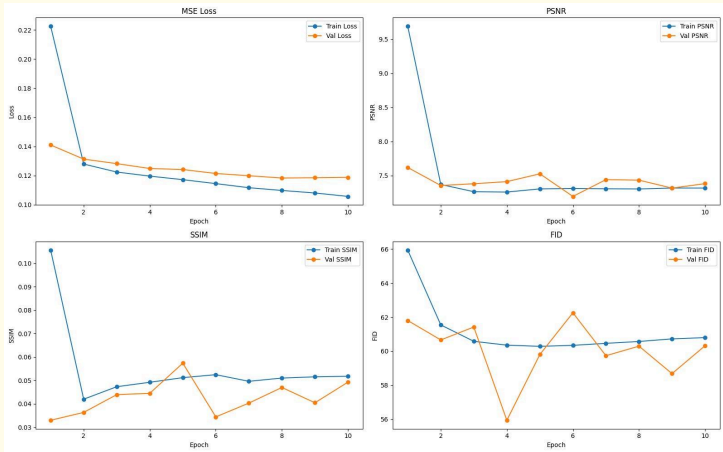


Table 2: Colorizer Performance

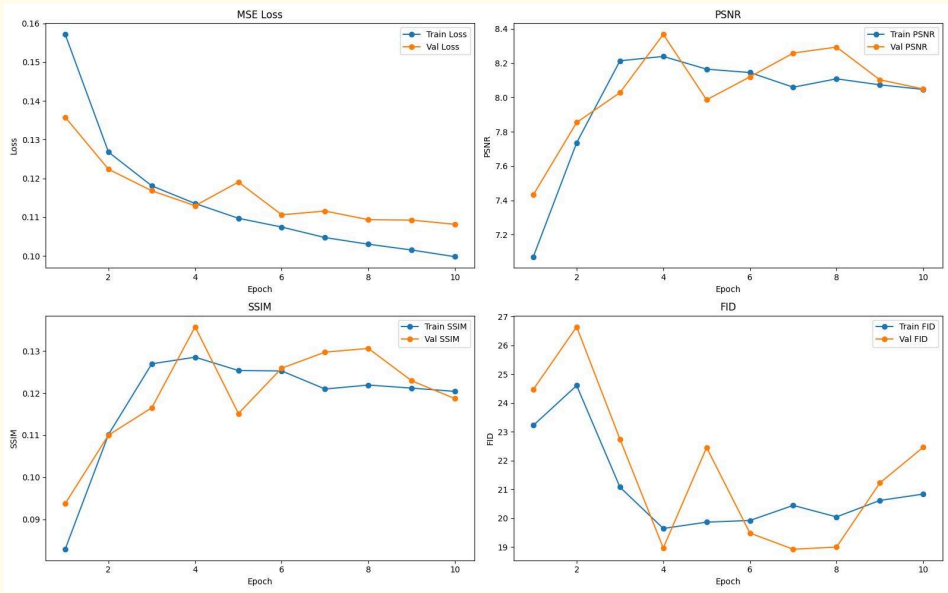
Performance Metric	Train Set	Val Set	Test Set	Test Set
	Final avg value per epoch	Final avg value per epoch	Average value	Average value
	FT Model	FT Model	Base Model	FT Model
MSE Loss	0.1058	0.1189	0.0405	0.0405
PSNR	7.32	7.38	4.10	4.10

SSIM	0.0518	0.0493	0.4475	0.4475
FID	60.80	60.32	223.20	223.20

6.2 Inpainting Quantitative Results

In addition to MSE Loss, PSNR, SSI, and FID was also measured for the fine-tuned (FT) Inpainting model pipeline performance. The LoRA rank for this model was increased to 20 for the StableDiffusionInpaintPipeline, resulting in 3,985,920 of 863,521,284 (0.46%) trainable parameters. This still appears to be not high enough as performance metrics appeared to roughly stabilize after 4 epochs of training as shown in Figure 7.

Figure 7: Inpainting performance metrics during training



This implies that training is slightly more effective than the colorizer, but still suggests a higher LoRA rank may be needed and also could explain the similar performance of the base model to the fine-tuned model on the test set shown in Table 3.

Table 3: Inpainting Performance

Performance Metric	Train Set	al Set	est Set	Test Set
	Final avg value per epoch	Final avg value per epoch	Average value	Average value
	FT Model	FT Model	Base Model	FT Model
MSE Loss	0.0998	0.1082	0.0327	0.0327
PSNR	8.05	8.05	15.66	15.66
SSIM	0.1204	0.1187	0.5337	0.5337
FID	20.84	22.46	186.76	186.76

7. Discussion

Our experiments show that diffusion models fine-tuned with Low-Rank Adaptation (LoRA) outperform both traditional image-processing baselines and the default Stable Diffusion pipelines in all four tasks. LoRA worked particularly well here because it allows for the adaptation of pre-trained weights at low computational cost while preserving the expressiveness of the full model. Successful deployment within a HuggingFace Space further supports the practicality and portability of our approach.

While full UNet fine-tuning did yield measurable improvements, notably for denoising, results were limited by resource constraints and short training schedules (≈ 5 epochs). The undertrained UNet in the super-resolution task resulted in oversmoothing and structural distortions, indicating that much longer training, more data, and refined schedules would be needed to have competitive performance.

8. Conclusion

The project successfully delivered:

- Four image enhancement tasks under one unified system
- Fine-tuned LoRA and full UNet training diffusion pipelines were created and tested.
- Denoising showed improved results even for short training epochs. Super resolution showed to produce scaled images and can be iterated on.
- A fully functional HuggingFace Space application
- A LoRA rank of 8 for the colorization pipeline allowed for UNet fine-tuning of only 0.18% trainable parameters that roughly stabilized performance metrics after only 2 epochs, so is determined to be too low for the training dataset size of 1000 grayscale and color images using the `StableDiffusionImg2ImgPipeline`.
- Increasing LoRA rank to 20 improved fine tuning performance metrics for a longer duration, but at 0.46% seems too low as Inpainting performance metrics roughly stabilized after 4 epochs using the `StableDiffusionInpaintPipeline`.

9. Future Work

Short-term:

- Add mask drawing tools to UI
- Add parameter sliders (strength, guidance scale, step count)
- Add metric evaluation (PSNR, SSIM, LPIPS)
- Debug UNet training pipeline further
- Add image augmentations in preprocessing
- Increase LoRA ranks
- Further investigation base pipeline VAE influence over fine tuned UNet
- Further hyper parameter adjustments to prompt,
- Incorporate natural language processing (NLP) tasks to generate image specific dynamic prompts to explain tasks with greater relevance to image details (Line et al., 2023)

Long-term:

- Full multi-GPU LoRA training
 - Triton or TensorRT deployment
 - Mobile app integration
 - Add face restoration, deblurring, old-photo restoration
-

Appendix

AI Declaration: AI (ChatGPT) was used to help explain some formulas and concepts but all technical work and conclusions are original.

Appendix A — Links

HuggingFace Space (UI)

<https://huggingface.co/spaces/dydsa/faai25>

Super Resolution Model

https://huggingface.co/dydsa/superres_unet

<https://huggingface.co/dydsa/superres>

Denoiser Model

https://huggingface.co/dydsa/denoiser_unet

<https://huggingface.co/dydsa/denoiser>

Colorizer LoRA Model

<https://huggingface.co/caliskate/inpainting-pipe-fine-tuned-aa1521>

Inpainting LoRA Model

<https://huggingface.co/caliskate/inpainting-LoRA-6-epochs>

GitHub Repo

<https://github.com/caliskate/ai-img-enhancement-webapp>

Appendix B — Background Cont...

Diffusion models represent the latest advancement, functioning as inherent denoisers through their noise-reversing generation process (Ho et al., 2020). Unlike CNNs that may oversimplify, diffusion models preserve fine details through iterative refinement (Kawar et al., 2022). Their generative capacity enables superior handling of ill-posed problems while maintaining natural image appearance. The ability to condition on various inputs—text, degraded images, or class labels—extends their applicability across diverse restoration scenarios, from low-light photography to medical imaging (Lugmayr et al., 2022).

2.5 Super-Resolution

Traditional interpolation introduces blur. GAN-based models like ESRGAN enhance sharpness but may hallucinate textures.

Diffusion-based super-resolution applies latent upscaling and iterative refinement, producing higher fidelity details.

Diffusion Models and Latent Diffusion

Diffusion models have emerged as a powerful class of generative models that synthesize high-quality images through iterative denoising. The process involves gradually adding noise to training data in a forward diffusion process, then learning to reverse this process through a neural network that predicts and removes noise step-by-step. While early diffusion models operated directly in pixel space, this approach proved computationally expensive, often requiring hundreds of GPU days for training and slow inference due to sequential evaluations.

Rombach et al. (2021) introduced **Latent Diffusion Models (LDMs)**, which address these computational challenges by applying the diffusion process in the latent space of pretrained autoencoders rather than directly on pixels. This approach achieves what the authors describe as a "near-optimal point between complexity reduction and detail preservation," significantly reducing computational requirements while maintaining high visual fidelity. The model architecture consists of three key components: a variational autoencoder (VAE) that compresses images into a lower-dimensional latent representation, a UNet that performs the diffusion process in this latent space, and cross-attention layers that enable flexible conditioning on inputs such as text prompts or bounding boxes. This framework has demonstrated state-of-the-art results across multiple tasks including image inpainting, unconditional generation, and super-resolution.

Stable Diffusion v1.5

The `runwayml/stable-diffusion-v1-5` model is an implementation of the Latent Diffusion Model architecture trained on large-scale image-text datasets. It operates on 512×512 images compressed to 64×64 latent representations with 4 channels (a compression factor

of 8), using a UNet with approximately 860 million parameters for the denoising process. The model employs CLIP's text encoder to process text prompts into conditioning vectors, enabling text-to-image generation through cross-attention mechanisms integrated throughout the UNet architecture.

Classifier-Free Guidance

A critical component of modern diffusion models' success is **Classifier-Free Guidance (CFG)**, introduced by Ho and Salimans (2022). This technique enables a balance between sample quality and diversity in conditional generation without requiring a separate classifier network. The method works by jointly training both conditional and unconditional versions of the diffusion model, then combining their score estimates during inference. Specifically, the guided prediction is computed as:

$$\epsilon_{\text{guided}} = \epsilon_{\text{uncond}} + w \times (\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$$

where ϵ_{uncond} is the unconditional noise prediction, ϵ_{cond} is the conditional prediction, and w is the guidance scale. When $w > 1$, the model emphasizes samples more closely aligned with the conditioning input, improving fidelity at the potential cost of diversity. This approach has become standard in text-to-image models like Stable Diffusion, where it enables users to control the trade-off between adherence to text prompts and output diversity.

Application to Image Restoration Tasks

While Stable Diffusion was originally designed for text-to-image generation, its latent diffusion architecture is well-suited for image-to-image translation tasks such as denoising and super-resolution. In these applications, the model can be conditioned on degraded input images rather than text prompts. For denoising, the model learns to map noisy inputs to clean outputs by operating in the `Img2Img` mode with appropriate conditioning. For super-resolution, specialized variants like `stabilityai/stable-diffusion-x4-upscaler` extend the architecture by concatenating low-resolution RGB inputs with noisy high-resolution latents, using noise level embeddings as additional class conditioning to guide the upscaling process. Fine-tuning these pretrained models on domain-specific datasets can further improve performance for specialized restoration tasks while requiring significantly less training data and computational resources than training from scratch.

Appendix C — Hardware & Environment

- HF GPU (A10G)
- Local: RTX 3090 (when available)
- Python 3.10, PyTorch 2.x, Diffusers 0.30
- Gradio 4.x
- Flask for backend

Appendix D — UNet Training Approach

Training Approach

Fine-tuning was performed using a dataset of paired (clean, noisy) images. Noisy images were generated using controlled Gaussian perturbations, consistent with the forward diffusion process described in Ho et al. (2020). During training, the U-Net was optimized to predict the residual noise ϵ added at each timestep t :

$$\mathcal{L} = \mathbb{E}_{x, \epsilon, t} [|\epsilon\theta(z_t, t, c) - \epsilon|^2]$$

where:

- (z_t) is the latent image after applying noise at timestep t
- (c) is the text conditioning embedding (we used null conditioning for pure denoising)
- $(\epsilon\theta)$ is the U-Net's predicted noise

Only the **U-Net weights** were set as trainable; the VAE, text encoder, and schedulers remained frozen to preserve stability and minimize compute cost. This is consistent with recent literature showing that U-Net-only fine-tuning is sufficient for downstream image transformation tasks (Ruiz et al., 2023; Meng et al., 2021).

Implementation Details

Training was performed in Google Colab using mixed-precision training (FP16) for efficiency.

Key hyperparameters:

- **Optimizer:** AdamW, learning rate $1e-4$
- **Batch size:** 4–8 depending on GPU memory
- **Noise scheduler:** DDIM

- **Training epochs:** 5- over ~3000 training pairs split into train/val/test
- **Gradient checkpointing:** Enabled to reduce VRAM usage
- **Checkpoint format:** Hugging Face Diffusers LoRA + full-model checkpoints

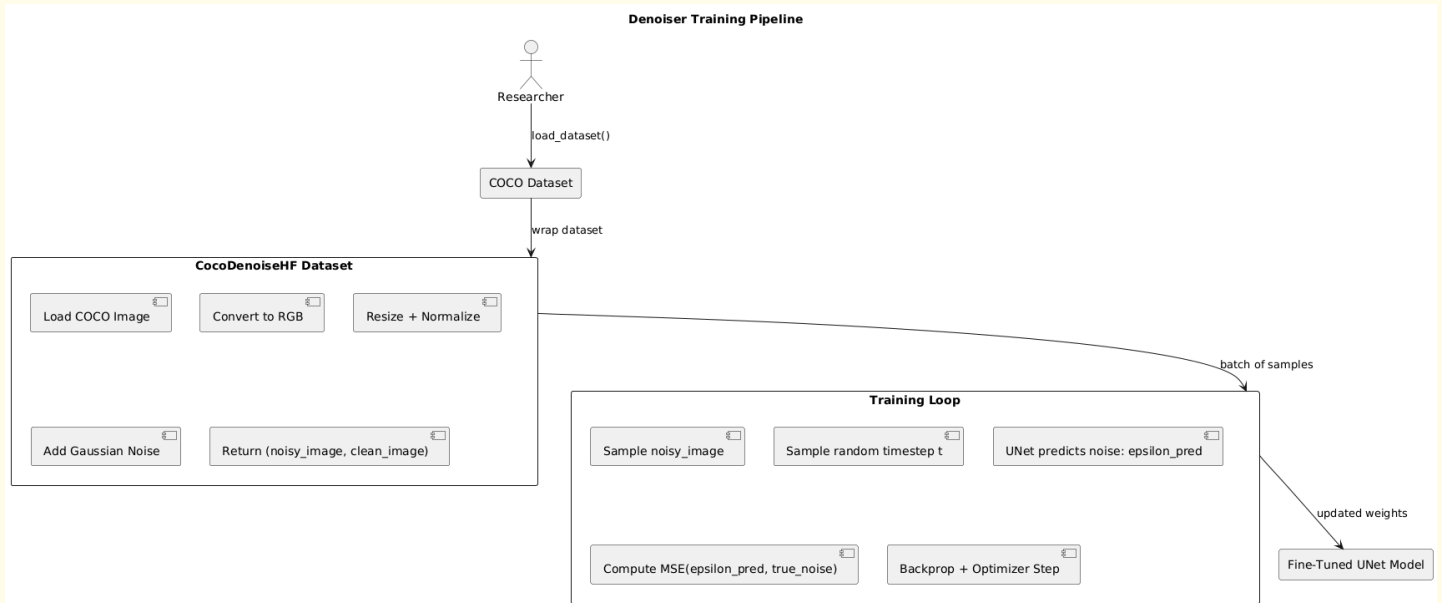
The resulting fine-tuned U-Net was uploaded to the Hugging Face Model Hub for easy versioning and integration with the Gradio-based inference app running in Hugging Face Spaces.

Performance Summary

Quantitative results demonstrated clear improvements compared to the pretrained Stable Diffusion v1.5 model:

- **PSNR:** 12.74 → **15.74**
- **SSIM:** 0.127 → **0.154**
- **LPIPS:** 0.7009 → **0.6604** (lower is better)

Figure 1: Denoiser and Super Resolution training architecture:



The denoiser training architecture fine-tunes a pretrained Stable Diffusion UNet to predict and remove Gaussian noise from corrupted images. The CocoDenoiseHF dataset synthetically adds noise ($\sigma = 0.3$) to COCO images, creating paired training samples. The training loop samples random timesteps and optimizes MSE loss between predicted and true noise. This approach leverages the diffusion model’s inherent denoising capability, the reverse diffusion process naturally removes noise corruption, while requiring minimal fine-tuning compared to training from scratch.

Super Resolution Implementation Details

Training Objective

During training, each low-resolution (LR) sample is paired with a high-resolution (HR) ground truth image. The LR image is encoded into the latent space using the frozen VAE, producing latents (z) . Gaussian noise (ϵ) is added according to the diffusion schedule:

$$z_t = \sqrt{\alpha_t} z + \sqrt{1 - \alpha_t} \epsilon$$

The U-Net is then trained to predict the injected noise:

$$\mathcal{L}_{SR} = \mathbb{E} [\|\epsilon\theta(z_t, t, c, x_{LR}) - \epsilon\|^2]$$

where

- (x_{LR}) is the low-resolution conditioning image,
- (c) is the text embedding (we use empty prompts for unconditional SR),
- $(\epsilon\theta)$ is the U-Net noise predictor.

This formulation follows the standard DDPM reconstruction loss (Ho et al., 2020) and is commonly used in diffusion-based super-resolution methods (Saharia et al., 2022).

Implementation Details

Only the **U-Net** was fine-tuned; the VAE, CLIP text encoder, tokenizer, and noise scheduler remained frozen. Training was conducted with:

- **Optimizer:** AdamW, learning rate $1e-5$
- **Batch size:** 4–8
- **Mixed precision:** FP16 via Hugging Face Accelerate
- **Gradient checkpointing:** Enabled
- **Dataset:** Paired LR–HR image dataset (HR images were downsampled on-the-fly to create LR counterparts)
- **Epochs:** 5
- **Scheduler:** Pretrained DDPM scheduler from the Upscale Pipeline

This mirrors recent work showing that U-Net-only fine-tuning is effective and computationally efficient for super-resolution tasks (Li et al., 2022; Saharia et al., 2022).

Figure 2 shows the high level pseudo code for training both the UNet for denoising and super resolution with minimal differences.

Figure 2: Pseudo Code For Full UNET re-training for denoiser and super resolution:

```
Initialize accelerator (mixed precision optional)
```

```
Load pretrained Stable Diffusion components:
```

- Variational Autoencoder (VAE)**
- CLIP text encoder**
- CLIP tokenizer**
- UNet denoiser network**
- DDPM Scheduler (diffusion noise scheduler)**

```
Freeze VAE parameters
```

```
Freeze text encoder parameters
```

```
Enable gradient checkpointing on UNet (optional)
```

```
Load training dataset of clean/noisy image pairs
```

```
Create DataLoader with batch size B
```

```
Initialize optimizer on UNet parameters
```


FOR each epoch:

FOR each batch of images x :

1. Encode images into latent space

latents = VAE.encode(x)

latents = latents * scaling_factor

2. Sample random diffusion timestep $t \in [0, T]$

t = random timestep for each sample in batch

3. Add noise according to diffusion process $q(z_t \mid z_0, t)$

$\epsilon \sim \text{Normal}(0, 1)$

noisy_latents = add_noise(latents, ϵ , t)

4. Prepare empty text conditioning

token_ids = tokenizer([""] * B)

text_embeddings = text_encoder(token_ids)

5. Predict noise using UNet $\epsilon_\theta(z_t, t, \text{text_embeddings})$

ϵ_{pred} = UNet(noisy_latents, t , encoder_hidden_states=text_embeddings)

6. Compute loss: noise prediction objective

loss = MSE(ϵ_{pred} , ϵ)

7. Backpropagate and update UNet weights

```
accelerator.backward(loss)
```

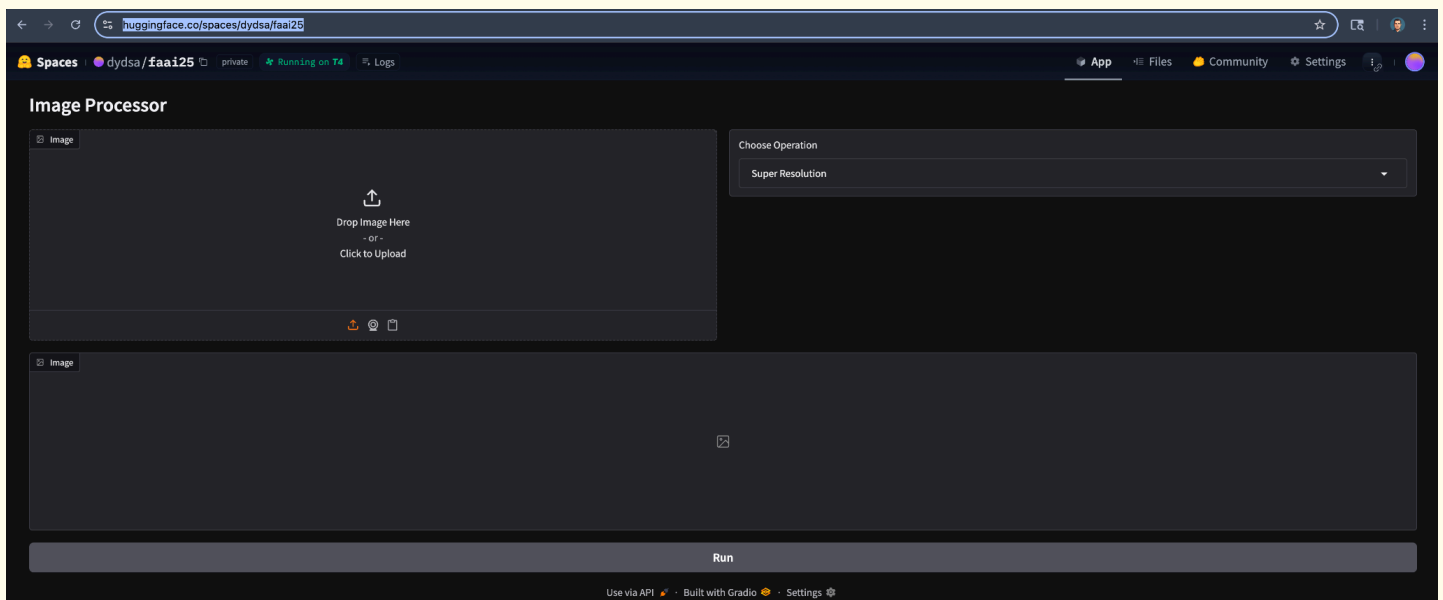
```
optimizer.step()
```

```
optimizer.zero_grad()
```

Save model checkpoint at end of epoch

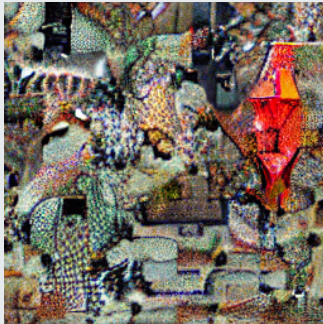
Save final fine-tuned UNet and diffusion scheduler

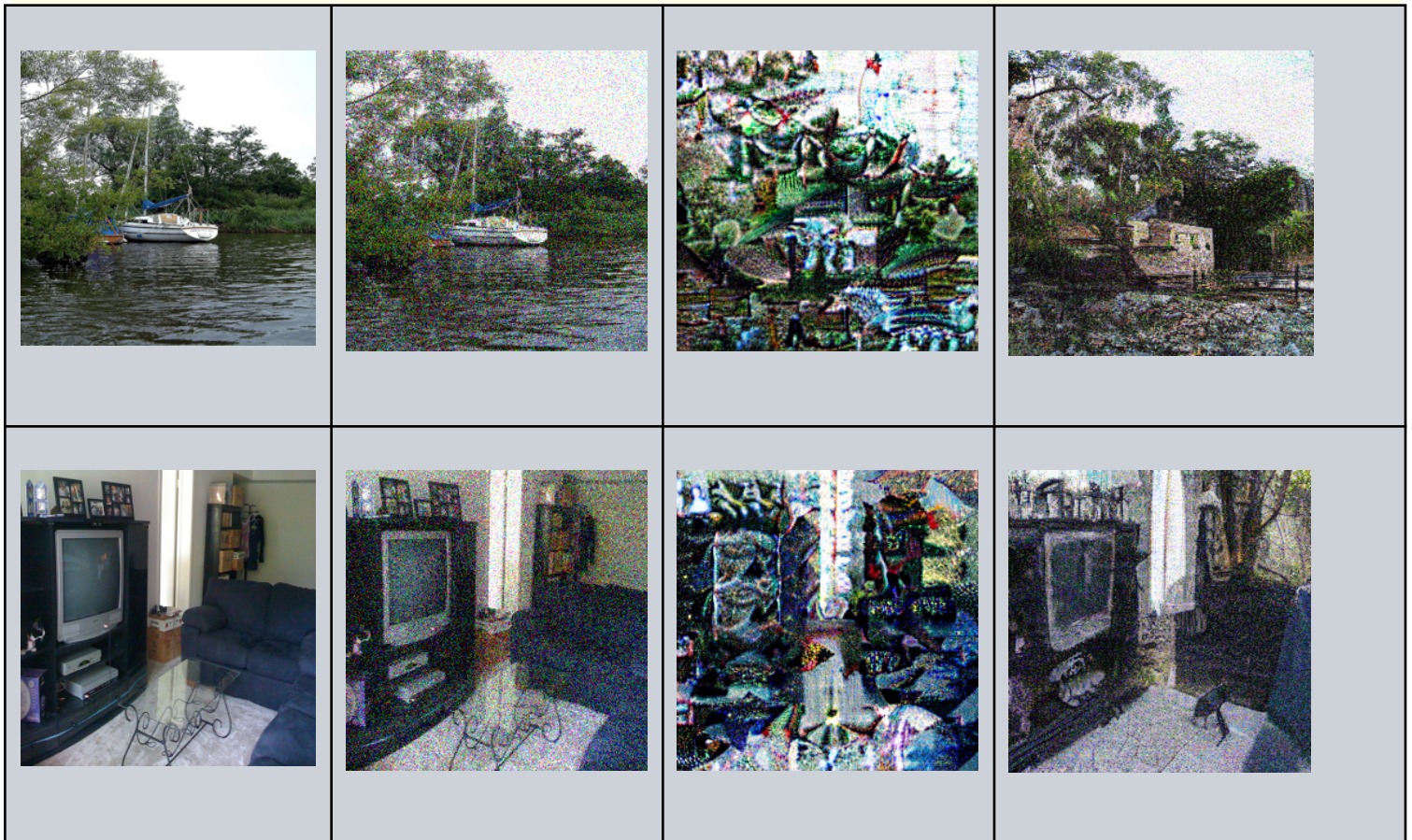
Appendix E App



Appendix F Denoiser and Super Resolution

Figure 3: Denoiser results

Original Image	Low Res Image	SD v1.5 Result	Denoiser Fine Tuned Result
			
			



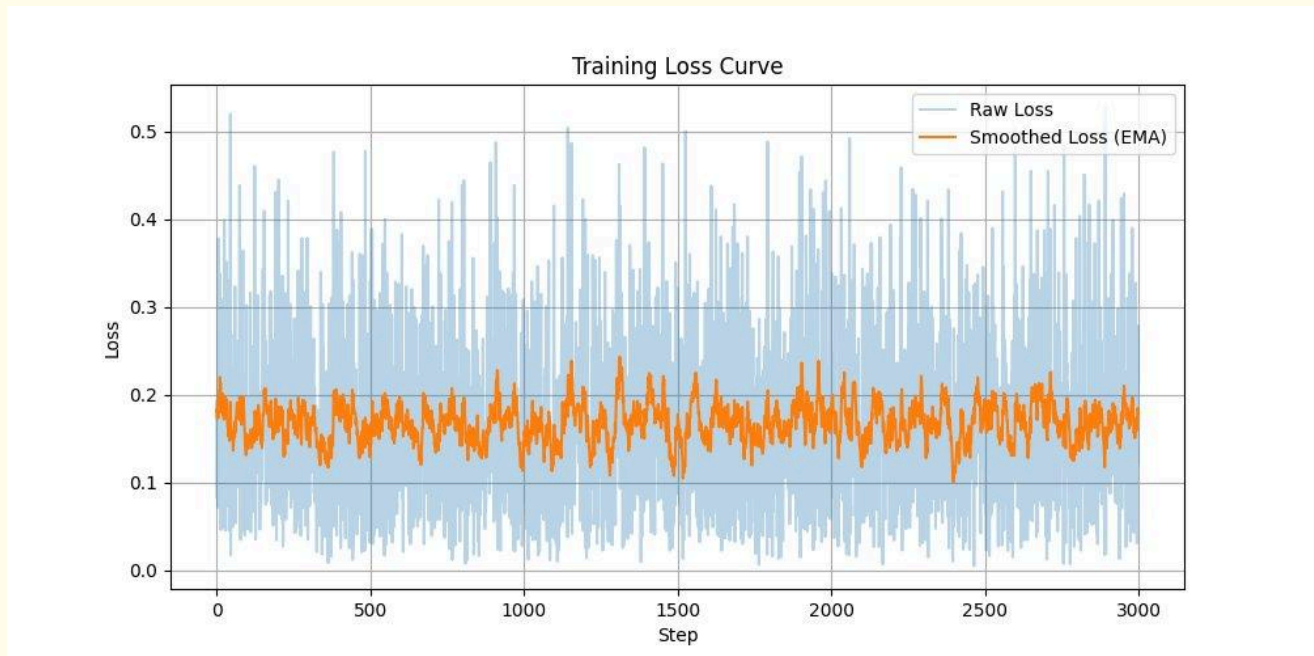
Results Cont..

Across all examples, the **low-resolution inputs** exhibit heavy speckle noise, significant loss of structure, and degraded color fidelity. When processed with the **pretrained SD v1.5 upscaler**, the images often become further distorted. Instead of reconstructing missing detail, the baseline model introduces hallucinated textures and inconsistent object boundaries. For instance, in the first row, the horses and background become visually implausible, with artifacts dominating the reconstruction. Similarly, in the second row, the fire hydrant and sand textures are replaced by unrelated patterns, reflecting the model's lack of specialization for noisy inputs.

In contrast, the **fine-tuned U-Net model** demonstrates consistently improved reconstruction. While not perfect, it restores coherent global structure, reduces hallucinations, and produces textures that better match the original image. In all rows, object contours—such as the horses, fire hydrant, and trees—are more faithfully preserved. Noise levels are substantially reduced, and overall image realism is improved.

These qualitative results indicate that domain-specific fine-tuning significantly enhances the upscaling performance of diffusion-based models although the results are not what we expected to see for both the SD v1.5 and fine tuned version. By training directly on low-resolution/noisy-to-high-resolution image pairs, the model learns degradation patterns that the pretrained base model fails to generalize to. We believe that there may be further improvements to be had (See improvements section) where more than 5-10 training epochs would yield better results. Due to time/cost constraints this further training is left as future work.

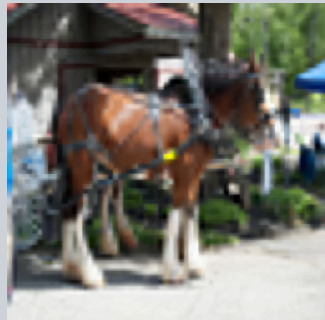
Figure 4 Training loss Super Resolution UNet training (EMA)



Super Resolution Results Cont...

Figure 5: Super resolution results for fully trained UNET component

Original Image	Low Res Image	SD v1.5 Result	SR Fine Tuned Result





Across all examples, SD v1.5 produces sharper edges and slightly improved textures, but it also tends to hallucinate patterns inconsistent with the original content. This is particularly noticeable in the horse image and the sand sculpture example, where noise-like artifacts appear after upscaling. Nevertheless, the baseline model generally maintains global structure.

In contrast, the **fine-tuned SR model performs significantly worse**, producing overly smoothed, cartoon-like reconstructions lacking recognizable detail. Object boundaries—such as the horse legs, the sand house, and the boat hull—are blurred into abstract shapes. This degradation is consistent across all samples, indicating that the model failed to properly learn the noise-to-detail mapping required for robust super-resolution.

The poor performance is likely due to **very limited training**: only *5 epochs* of UNet training and approximately *25 total iterations* per epoch. This is far below the scale typically required for SR fine-tuning, where tens of thousands of updates are common. Because of time and resource constraints, further training and hyperparameter tuning were left to future work.

We expect that a longer training schedule, improved dataset preparation, and possibly LoRA-based adaptation would yield substantially better results.

Appendix G Metrics

Stable Diffusion (and many other diffusion-based image generators) are built on a U-Net denoiser that is trained inside a GAN-like adversarial framework (the “diffusion-prior” can be thought of as a generator and the noise schedule can be considered a discriminator-style loss).

Because the model is learning to create photorealistic, high-fidelity images from pure noise, we need objective ways to measure three things simultaneously:

Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) are complementary image-quality metrics.

PSNR quantifies pixel-wise fidelity, higher values mean the output matches the ground-truth more closely, making it useful for super-resolution and denoising. SSIM evaluates perceptual structure, comparing textures, edges, and luminance, and aligns better with human vision than PSNR. LPIPS goes further by measuring similarity in deep-network feature space, capturing edges, textures, and semantic content, thus reflecting human judgments more accurately than traditional pixel-based scores.

PSNR — Peak Signal-to-Noise Ratio (pixel-wise fidelity)

- Measures how closely the output image matches the ground truth at the pixel level.

- Higher = better.
- Good for super-resolution and denoising tasks.
- Complementary to SSIM.

SSIM — Structural Similarity Index (perceptual structure)

- Measures similarity in textures, edges, and luminance.
- Higher = better.
- Aligns better with human perception than PSNR.

LPIPS = Learned Perceptual Image Patch Similarity

It is a metric for measuring how *visually similar* two images are in a way that aligns with human perception, rather than raw pixel differences.

It was introduced by *Zhang et al., 2018*.

Traditional metrics like:

- **L2 / MSE**
- **PSNR**
- **SSIM**

mainly compare **pixels**.

But small pixel shifts or texture changes can cause huge errors even when images look nearly or almost identical to humans.

LPIPS solves this by measuring similarity in **deep neural feature space**, which captures:

- edges

- textures
- structures
- semantic content

This aligns much better with human judgment.

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